EXPERIMENT 8

Configuring an Event Registration & Participant Management Application with Role-Based Access Control (RBAC) in Zoho Creator

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Title

Designand Implementation of an Event Registration & Participant Management Application with User Permissions, Roles, Role Hierarchy and Data Sharing in Zoho Creator.

Aim

Tobuild an event-tracking application in Zoho Creator using a structured 13-field registration form (***Event_Registration_Form***) and report (***Event_Registration_Report***), and to implement granular role-based access control through a three-level role hierarchy and a custom Read Only data-sharing rule.

Objectives

- Todesign***Event_Registration_Form*** with 13 Basic Fields grouped into four layout panels: Identity, Contact, Preferences and Scheduling.
- To create ***Event_Registration_Report*** and a dashboard showing registrations, capacity achieved and participant distribution.
- To define the role hierarchy ***CEO → Organizing Committee Head → Track Manager***.
- To configure a data-sharing rule from ***Organizing Committee Head*** to ***Track Manager*** with ***Read Only*** access.
- To understand how RBAC separates data ownership from data visibility.

Software / Tools Required

- Zoho Creator(web-basedlow-code application development platform)
- Event Registration application created in Zoho Creator (created on Sep 28, 2026)
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

Introduction / Theory

User Management in Zoho Creator

User management decides who can access an application and what each person may see and do within it. Users are added to an application and associated with permission sets (profiles) and roles; together these control access to forms, reports, pages and records. In the Settings area, the User Permissions section groups these controls under three tabs: ***Permissions***, ***Roles*** and ***Data Sharing***.

Role-Based Access Control (RBAC)

RBAC is an access-control model in which rights are assigned to roles instead of individual users, and users acquire rights through the roles they hold. It simplifies administration, supports the principle of least privilege, improves data security and keeps the access policy consistent and easy to audit.

Key Concepts

Concept	*Description*
Users	Individuals added to the application who sign in to work with its forms, reports and data.

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Concept	*Description*
Permissions	Named permission sets (profiles) defining what a user may do with components. This application lists ***Committee_Organizer_Access***, ***Read*** and ***Write***.
Roles	Positions in the organisational structure (CEO, Organizing Committee Head, Track Manager) that determine a user's place in the hierarchy and the visibility of records.
Role Hierarchy	A tree in which each role reports to another. A higher role can generally view the records of roles beneath it; the reverse does not apply.
Data Sharing	Share rules that extend record visibility beyond the hierarchy, from one role or user to another, with a defined level of access (Read Only or Read/Write).

Application Specifications

Parameter	*Detail*
Application Name	Event Registration & Participant Management
Form Name	Event_Registration_Form
Report Name	Event_Registration_Report
Form Architecture	13 Basic Fields in 4 layout panels (Identity, Contact, Preferences,
Role Hierarchy	Scheduling) CEO → Organizing Committee Head → Track Manager
Custom Sharing Rule	Organizing Committee Head → Track Manager (Read Only)

Procedure

1. Signinto Zoho Creator. The home page lists the available applications. Locate the ***Event Registration*** application (Figure 8.1).

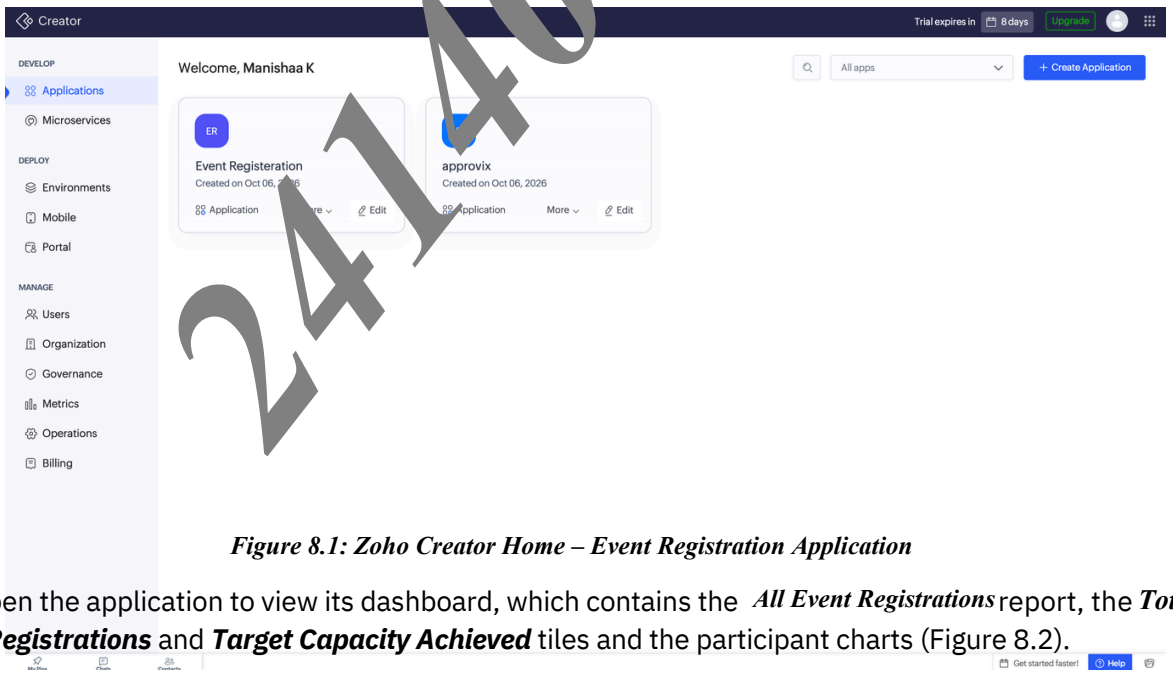


Figure 8.1: Zoho Creator Home – Event Registration Application

2. Open the application to view its dashboard, which contains the ***All Event Registrations*** report, the ***Total Registrations*** and ***Target Capacity Achieved*** tiles and the participant charts (Figure 8.2).

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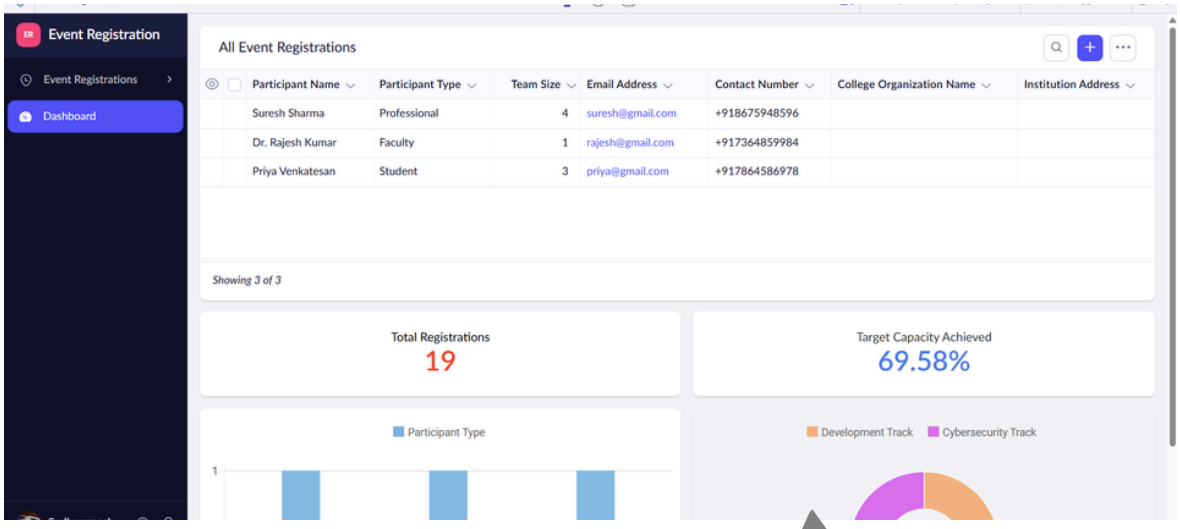


Figure 8.2: Event Registration Application – Access Mode

3. Enter Edit mode by selecting ***Edit*** on the application card (Figure 8.1) or ***Edit this application*** at the top-right (Figure 8.2).
4. Select the ***Settings*** tab from the top navigation bar (Design, Workflow, Settings).
5. Under the PERMISSIONS group, select ***User Permissions***. The Users page is displayed with the tabs Permissions, Roles and Data Sharing.
6. Study the ***Permissions*** tab. Three permission sets are listed: ***Committee_Organizer_Access***, ***Read*** and ***Write***. An ***Add Permission*** button is also provided (Figure 8.3).

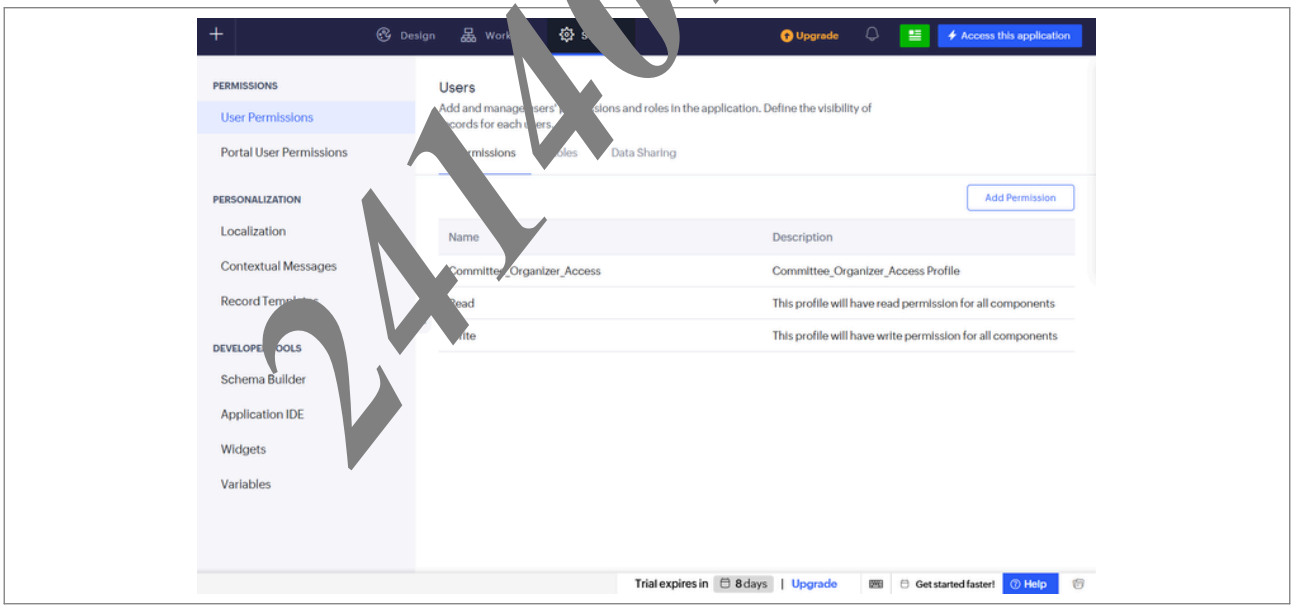


Figure 8.3: User Permissions – Permissions Section

7. Select the ***Roles*** tab and arrange the role tree as ***CEO → Organizing Committee Head → Track Manager***. The Expand All, Collapse All, Manage Role Hierarchy and Add Role options are available (Figure 8.4).

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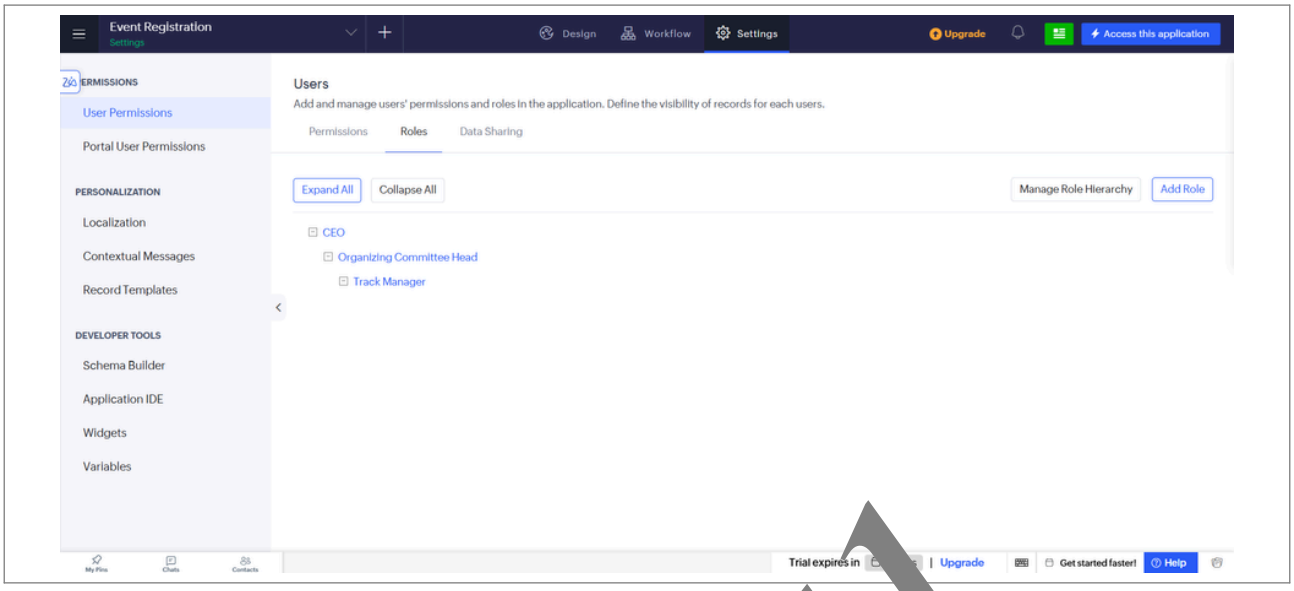


Figure 8.4: Roles and Role Hierarchy

8. Select the ***Data Sharing*** tab and choose *New ShareRule*. Enter Name: ***Event_Registration_Form***; Form: ***Event Registration***; Records shared from:***Roles***.The dropdown offers Roles, Roles and Subordinates, and Users (Figure 8.5).

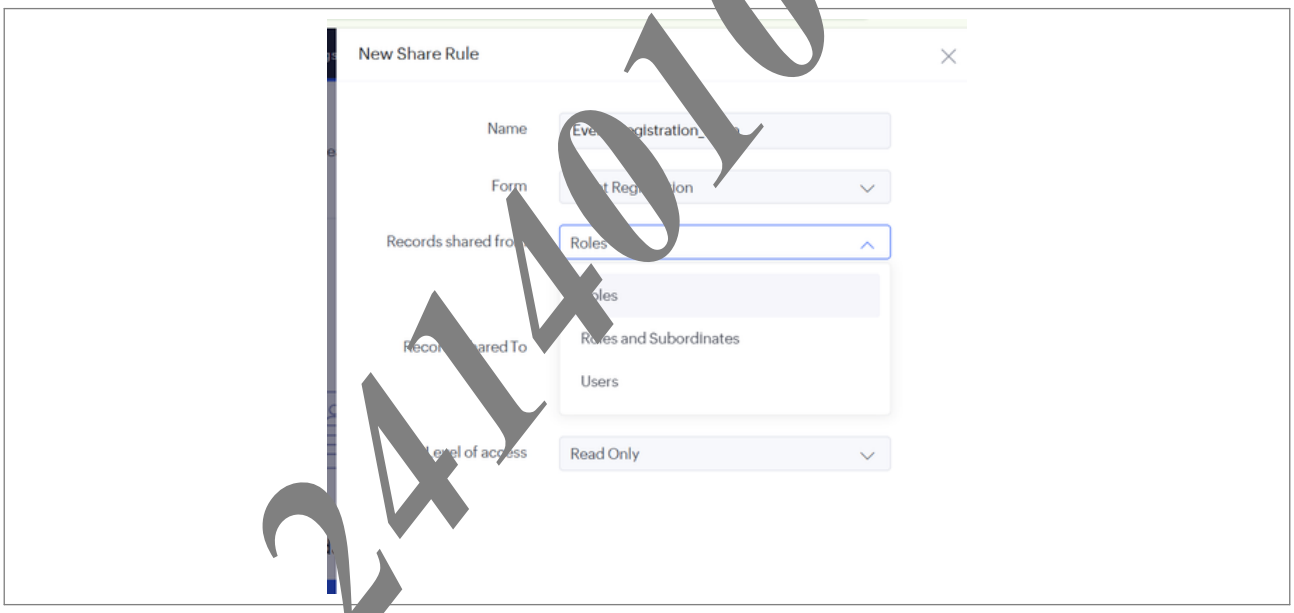


Figure 8.5: New Share Rule – Rule Configuration

9. Set Records shared from→ Roles → ***Organizing Committee Head***; Records shared To → Roles → ***Track Manager***; Level of access→ ***Read Only***.

10. Select ***Save***. The rule appears in the Data Sharing list and can be reopened with Edit Sharing Rule (Figure 8.6).

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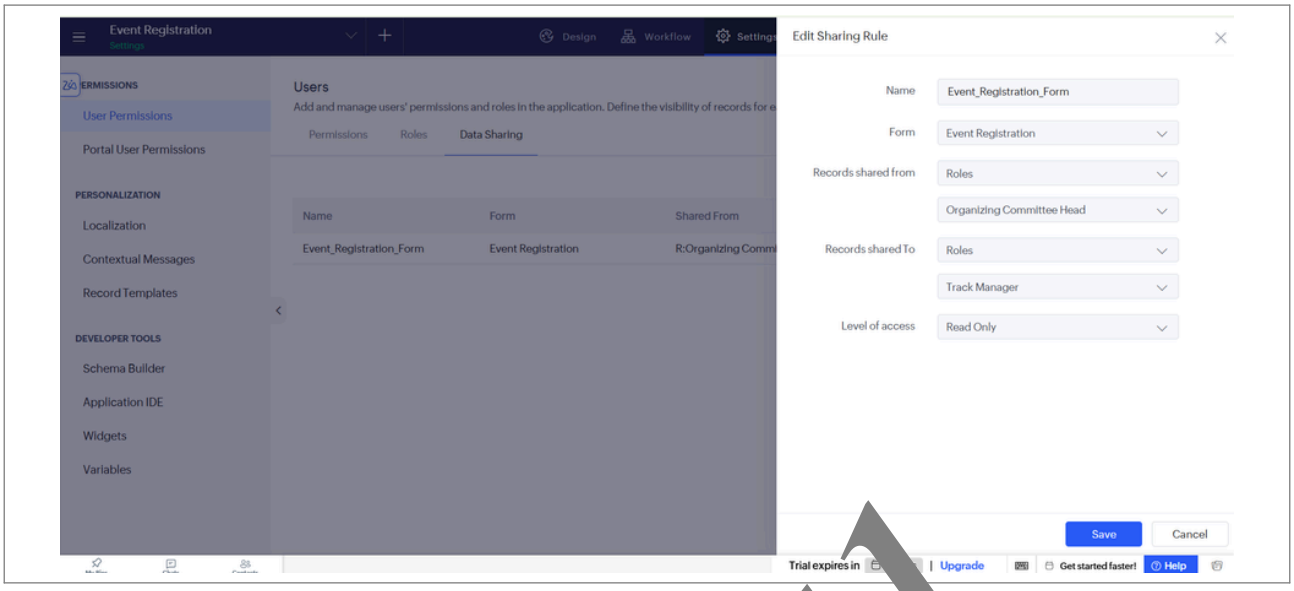


Figure 8.6: Configured Data Sharing Rule (Organizing Committee Head → Track Manager, Read Only)

Data Sharing

The application follows a layered model: ***Form*** (Event_Registration_Form) → ***Report*** (Event_Registration_Report) → ***Dashboard*** → ***Security Layer*** (Permissions, Roles, Data Sharing). The role hierarchy runs from the ***CEO*** (overall oversight) to the ***Organizing Committee Head*** (event-wide management) and then the ***Track Manager*** (track-level operations). By default the hierarchy gives only upward visibility: higher roles see the records of the roles below them, but not the reverse.

The custom data-sharing rule shares Event_Registration_Form records from the ***Organizing Committee Head*** role to the ***Track Manager*** role at ***Read Only*** level. Read Only preserves enterprise security because Track Managers can view the participant details they need for planning (counts, tracks, contact information) but cannot edit or delete them, so the records held by the Organizing Committee Head remain the single source truth and cannot be altered downstream, whether by accident or deliberately. It enforces least privilege by granting visibility without modification rights, keeps the scope directional (Head → Manager) so access never flows upward or sideways, and leaves a single named rule in the Data Sharing list that documents who can see what, and at what level, making the policy easy to audit.

RBAC Working

Level	***Role***	***Intended Function***
1 (Top)	CEO	Overall authority; able to view the records of roles reporting to it.
2.	Organizing Committee	Manages the whole event; owns the master registration records.
3.	Head Track Manager	Manages an individual track; views shared registrations read-only.

Users in the Track Manager role work with registration records according to their permission set.

Because Track Manager reports to the Organizing Committee Head, who reports to the CEO, higher roles can view the records of the roles below them.

- Lower roles do not automatically gain access to higher-role records; the Read Only share rule grants it in a controlled way.
- The permission sets (Committee_Organizer_Access, Read, Write) further limit the operations each user may carry out.

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Result

Thefollowing were successfully carried out in the Event Registration application:

- The Event Registration application, with Event_Registration_Form and Event_Registration_Report, was opened and its dashboard viewed.
- The permission sets Committee_Organizer_Access, Read and Write were studied in User Permissions.
- The role hierarchy CEO → Organizing Committee Head → Track Manager was created.
- A data-sharing rule from Organizing Committee Head to Track Manager with Read Only access was configured and saved.

Conclusion

Theexperiment demonstrated how permissions, roles, role hierarchy and data sharing together form a role-based access control mechanism in Zoho Creator. Structured registration with typed fields replaces unstructured sign-ups (spreadsheets, chat messages and free-text emails) with clean, consistent and analysable data. The three-level hierarchy and the directional Read Only share rule keep information channels isolated: Track Managers see only what they need and cannot alter master records, while the CEO and Organizing Committee Head retain oversight. The result is a secure, auditable and scalable event-management framework.

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